

Name: _____

Date: _____

PLACE VALUE AND ORDERING INTEGERS, (ORDERING POSITIVE AND NEGATIVE INTEGERS)

LO: I can order and compare positive and negative integers using a number line.

Ordering integers

Integers are whole numbers including negatives, zero, and positives. On a number line, numbers increase from left to right.

Key rule: A negative number is always less than a positive number. The further left on the number line, the smaller the value.

Quick examples

●	$-5 < -2 < 0 < 3 < 7$	ordered smallest to largest
●	-10 is less than -3	further from zero = smaller
●	0 is greater than any negative number	zero is not negative
●	$-5 > -2$ because $5 > 2$	ignore the sign is wrong

Key idea

Picture a number line: numbers to the right are always bigger. Negative numbers are to the left of zero.

$$-8 < -3 < 0 < 4 < 9$$

Left = smaller, Right = bigger

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - ordering negatives

Put these in order from smallest to largest: -3, 5, -7, 2, -1

Most negative = smallest: -7

Then -3, -1, 2, 5

-7, -3, -1, 2, 5

-7, -3, -1, 2, 5

The more negative a number, the smaller it is.

Example 2 - comparing two integers

Which is greater: -4 or -9?

-4 is closer to zero than -9

On the number line, -4 is to the right of -9

$$-4 > -9$$

-4 is greater

Closer to zero means bigger for negative numbers.

Example 3 - using inequality symbols

Insert < or >: $-6 \underline{\hspace{1cm}} -2$

On the number line, -6 is to the left of -2

Left means smaller

$$-6 < -2$$

-6 < -2

The open end of < points to the bigger number.

Example 4 - temperature context

Order these temperatures from coldest to warmest: 3°C , -5°C , -1°C , 8°C , -8°C

Most negative = coldest: -8°C

Then -5°C , -1°C , 3°C , 8°C

Coldest to warmest: -8, -5, -1, 3, 8

-8°C , -5°C , -1°C , 3°C , 8°C

Same as ordering on a number line.

YOUR TURN – PRACTICE (deliberate practice)

1. Ordering integers

Put in order from smallest to largest.

- a) 3, -2, 5, -4, 0
- b) -7, -1, 4, -3, 6
- c) 8, -8, 3, -3, 0
- d) -10, -5, -15, 2, -1

2. Comparing integers

Insert $<$ or $>$ between each pair.

- a) -3 ___ 2
- b) -5 ___ -8
- c) 0 ___ -4
- d) -12 ___ -7

3. Ordering with larger numbers

Order from smallest to largest.

- a) -25, -30, 15, -10, 20
- b) 100, -200, 50, -50, -100
- c) -1, -11, -111, 1, 11
- d) 0, -1000, 500, -500, 1

4. Real-world contexts

Order from smallest to largest.

- a) Temperatures: -3°C , 7°C , -12°C , 0°C , 4°C
- b) Bank balance: $-\text{£}50$, $\text{£}20$, $-\text{£}15$, $\text{£}0$, $-\text{£}80$
- c) Sea level (m): -40, 120, -120, 5, -5
- d) Floor levels: -2, 3, -1, 0, 5

5. Mixed ordering

Answer each question.

- a) Which is smaller: -99 or -100?
- b) Order: 4, -4, 40, -40, 0
- c) Find a number between -5 and -3
- d) How many integers between -3 and 3 (not including -3 and 3)?
- e) Order: -2.5, -3, 1, -1.5, 0.5

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) -5 is less than -2 ☐

Correct answer: _____

Reason: _____

b) -5 > -2 because $5 > 2$ ☐

Correct answer: _____

Reason: _____

c) 0 is greater than any negative number ☐

Correct answer: _____

Reason: _____

d) -100 is greater than -1 ☐

Correct answer: _____

Reason: _____

e) On a number line, right means bigger ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Five cities have temperatures: Moscow -15°C , London 4°C , Reykjavik -3°C , Dubai 28°C , Oslo -8°C . Order them coldest to warmest. Find the difference between the warmest and coldest.

Expression: _____

Simplified: _____

b) A lift starts at floor -2 (underground car park). It goes up 5 floors, then down 3 floors, then down 4 floors. Which floor is it on? Write the floor numbers visited in order from lowest to highest.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

